

KIDS' PLAYROOM BUILD

Corner Loft · Rope Bridge · Ranger's Outpost Playhouse

CONSTRUCTION DOCUMENT SET · dimensional lumber + stock hardware only · generated from the parametric Blender model

ROOM 13'-6" x 13'-5" · CEILING 8'-0" · PLATFORM 3'-8" · GUARD 3'-0"

READ FIRST — THREE SPECS IN THE BRIEF ARE NOT BUILDABLE AS WRITTEN

1. A 36-38" hobbit door will not fit under a 44" platform

BRIEF SAID

R.O. 24-26" W x 36-38" H

THIS SET USES

R.O. 26" W x 32" H

44" deck - 3/4" plywood - 5-1/2" joist = 37-3/4" of wall, floor to underside of framing.
Take out the 1-1/2" bottom plate, a 1-1/2" flat header, a 2-3/4" cripple and a 1-1/2" top plate and 32" is what is left.
A 32" door is still right for a 3-6 year old. The semicircular arch on top reads taller than it measures.

2. The knee wall studs are 34-3/4", not 40-42"

BRIEF SAID

studs ~40-42" tall

THIS SET USES

studs 34-3/4"

Same arithmetic. The wall dies into the underside of the joists at 37-3/4", and two plates eat 3" of that.
Cut one, hold it in place, and confirm before you cut the other fifteen.

3. The bridge span is 54", not 42"

BRIEF SAID

span ~42"

THIS SET USES

clear span 54"

162" wall - 60" loft - 48" playhouse = 54" of clear air between them.
Both structures are in corners, so neither can slide toward the other to close the gap.
Order the net for 54". Expect 6-8" of sag at that span, which is why the plan set carries a top rope AND a mid rope.

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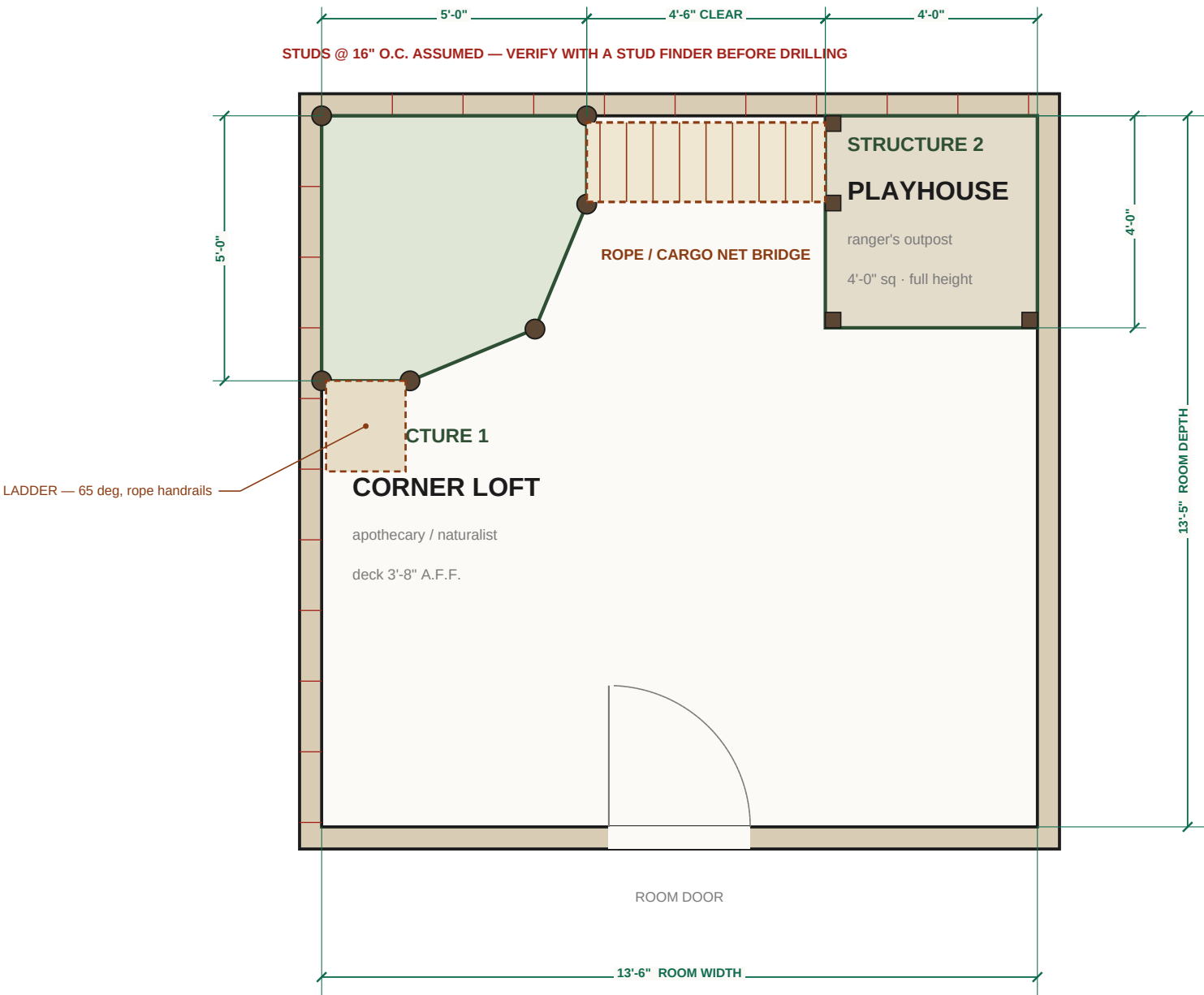
AT A GLANCE

Loft deck area	21.75 sq ft (60" x 60" envelope, hexagonal)
Loft deck shape	6-sided · angles 90/90/157.5/135/157.5/90
Miter angles required	11-1/4 deg and 22-1/2 deg — both saw detents
Posts	6 at the loft (5 tall + 1 corner stub), 4 at the playhouse
Bridge clear span	4'-6" · net 1'-6" wide
Guard height	3'-0" above deck (top of the 32-36" range)
Total 4x4 PT	~65 lf Total 2x4: ~127 lf 2x6: ~46 lf
Plywood	3 sheets 3/4" (2 structural, 1 skin/doors)
Est. material cost	\$1,250 - \$1,350 see sheet A9

OVERALL ROOM PLAN

structure locations · wall anchor zones · stud layout

A1



LOAD PATH

The wall ledgers carry the real load.
Posts are secondary / aesthetic.

LOFT: two 2x6 PT ledgers, one on the north wall and one on the west wall, 3/8" x 4" lag + washer into EVERY stud.

PLAYHOUSE: north and east posts lag-bolted directly to wall studs.

BRIDGE: both ends through-bolted into 2x6 ledgers that are themselves lagged into a minimum of 2 studs.
Eye BOLTS with washer + nylock nut.
Never screw eyes.

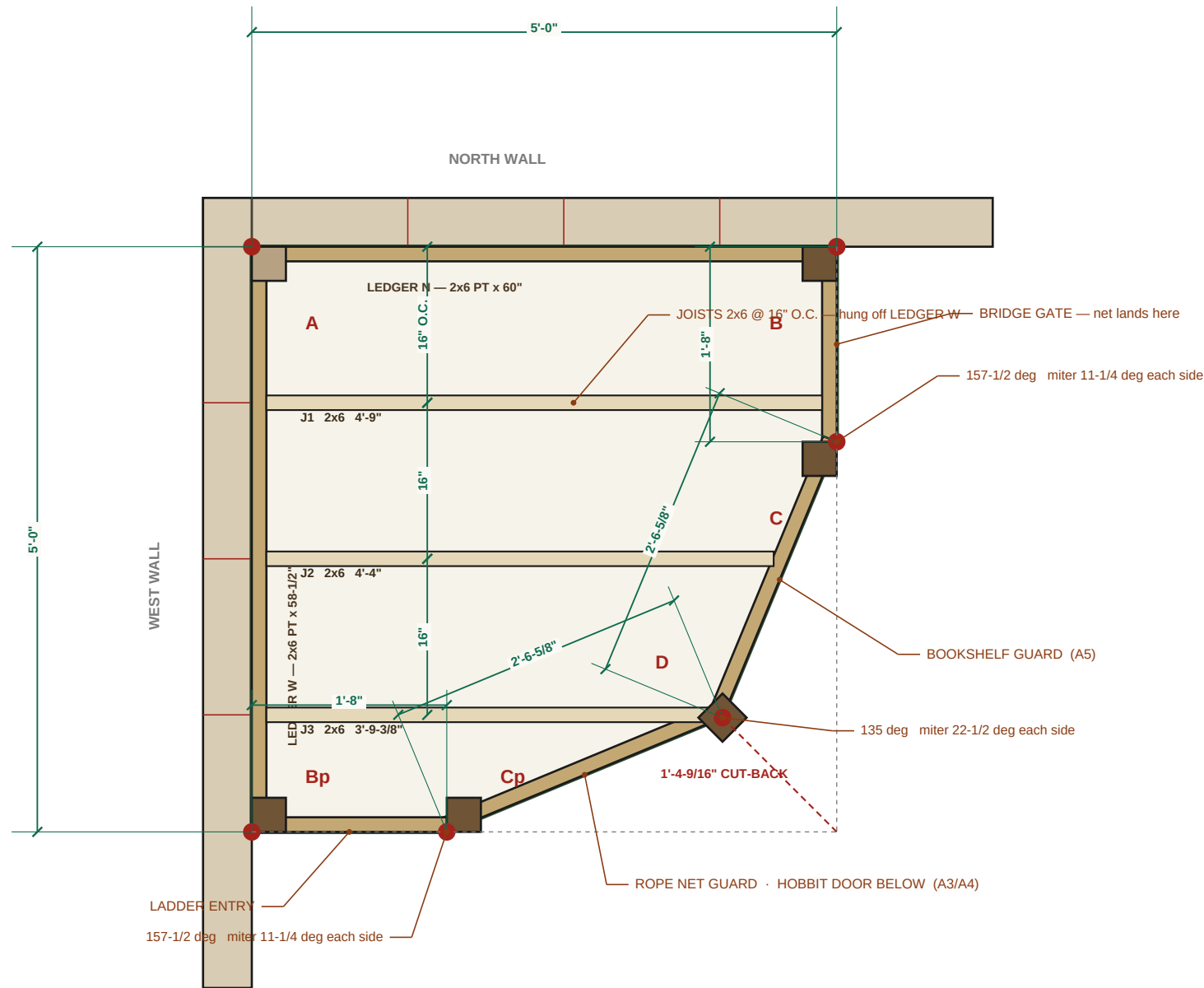
VERIFY STUD SPACING ON SITE.

16" O.C. is assumed throughout; if your studs are 24" O.C. or irregular, shift ledger bolt locations to land on wood and add a bolt — do not rely on drywall anchors anywhere in this set.

LOFT — DECK FRAMING PLAN

hexagon geometry · joist layout · post schedule

A2



HEXAGON GEOMETRY — RESOLVED

VERTEX	u (in)	v (in)	POST CUT
A	0.000	0.000	4x4 x 37-3/4"
B	60.000	0.000	4x4 x 80"
C	60.000	20.000	4x4 x 80"
D	48.284	48.284	4x4 x 80"
Cp	20.000	60.000	4x4 x 80"
Bp	0.000	60.000	4x4 x 80"

WHY THIS SHAPE

A 45-degree chamfer of one corner gives a PENTAGON, not a hexagon — the two cut edges come out collinear, so the middle vertex is not a corner at all.

A true 6-sided deck inside a 60 x 60 corner needs two 22-1/2 degree joints flanking one 45 degree joint. Interior angles: 90 / 90 / 157-1/2 / 135 / 157-1/2 / 90. They sum to 720, as a hexagon must.

Every miter is therefore 11-1/4 or 22-1/2 degrees. Both are detents on any 10-inch miter saw. Nothing in this deck needs a custom angle or a bevel gauge.

DECK AREA 21.75 SQ FT

Both 60-inch wall runs stay full length, so both ledgers get maximum stud engagement. That is the whole point of cutting the outer corner instead of the wall corners.

JOIST NOTES

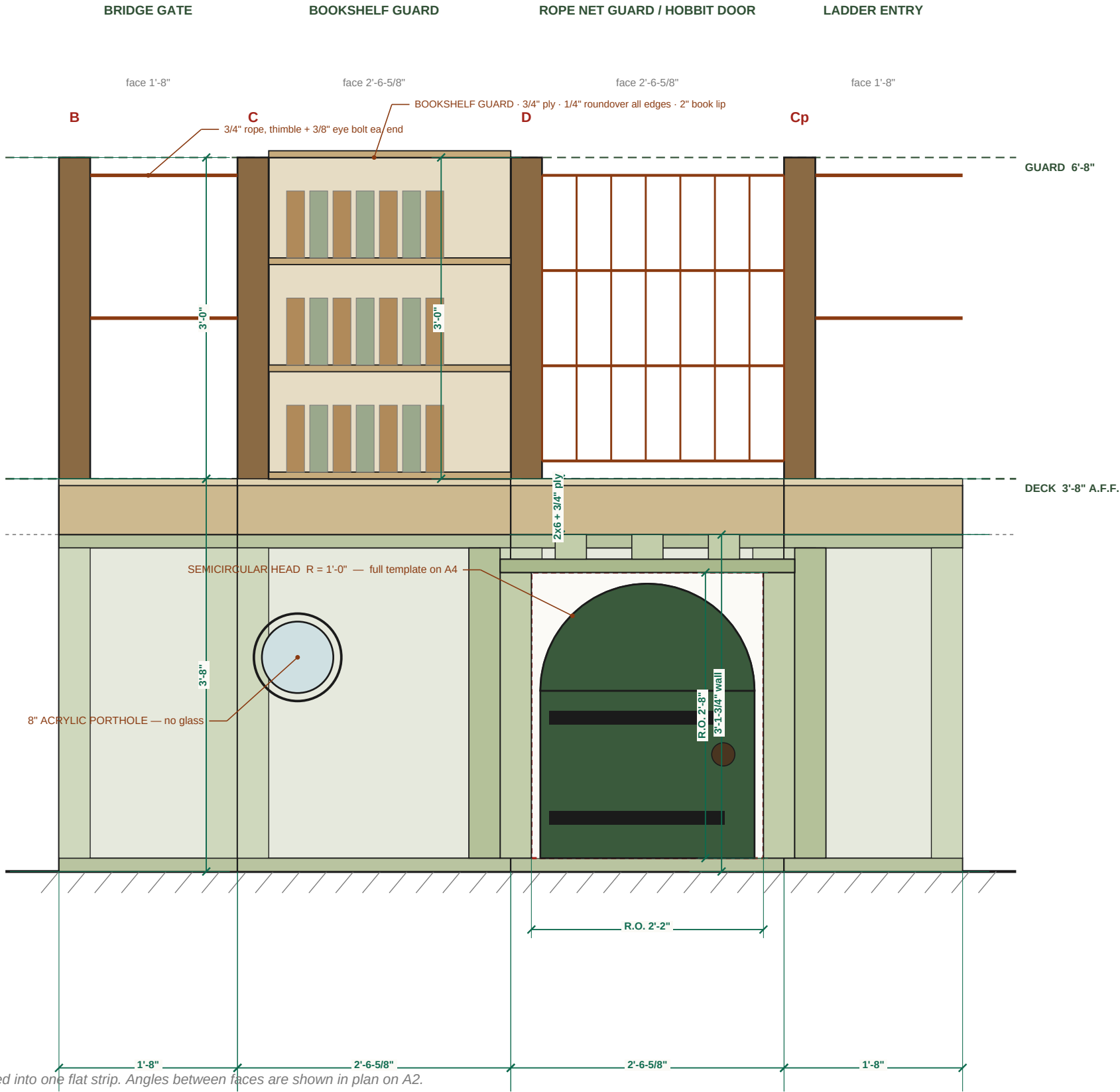
J1 is square both ends — Simpson LUS26 hangers. J2 and J3 die into an angled rim. Use Simpson LSU26 adjustable skewable U hangers (Internet # 100375129), field-bent to the angle shown. Alternatively bevel the joist end and add a 2x6 cleat underneath. Do not toe-nail either one.

POST D sits at 45 degrees so its faces meet both angled rims squarely. Cut its notches before standing.

LOFT — DEVELOPED ELEVATION

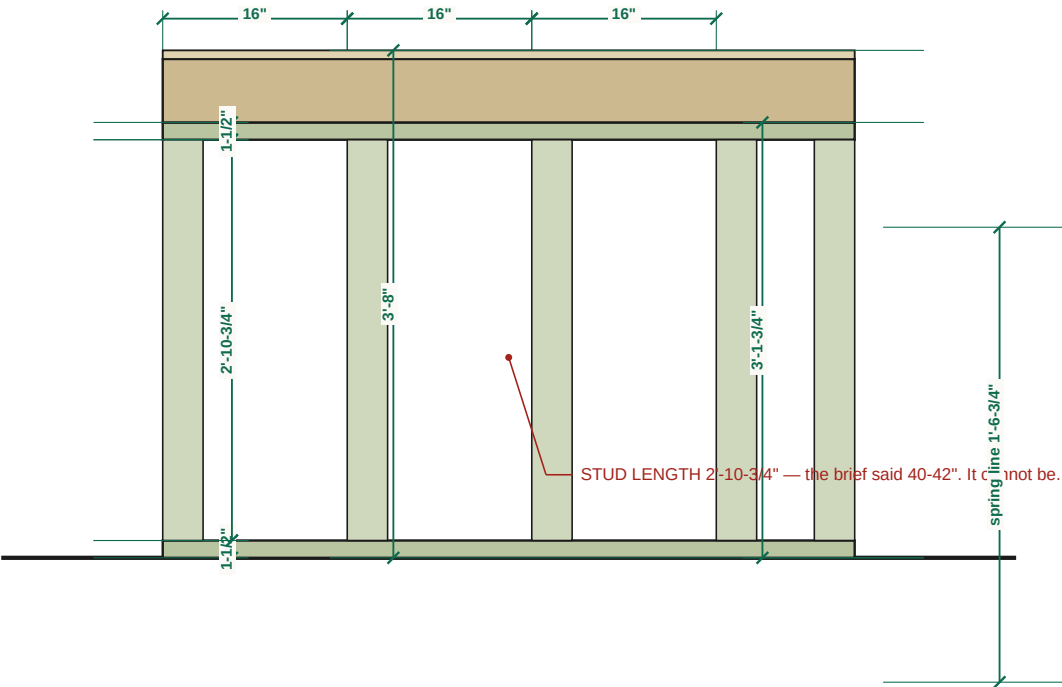
all four open faces unrolled flat

A3



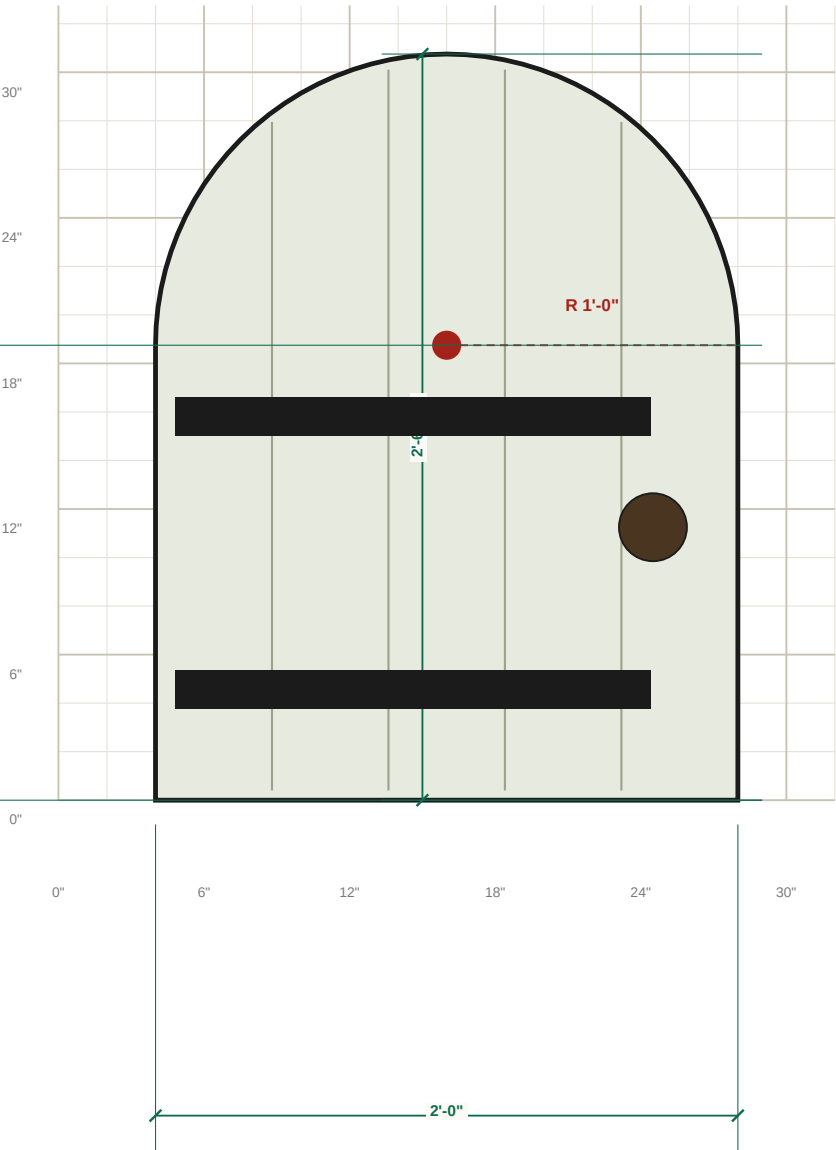
DEVELOPED ELEVATION — the four open faces of the loft unrolled into one flat strip. Angles between faces are shown in plan on A2.

KNEE WALL SECTION — typical



HOBBIT DOOR — FULL TEMPLATE

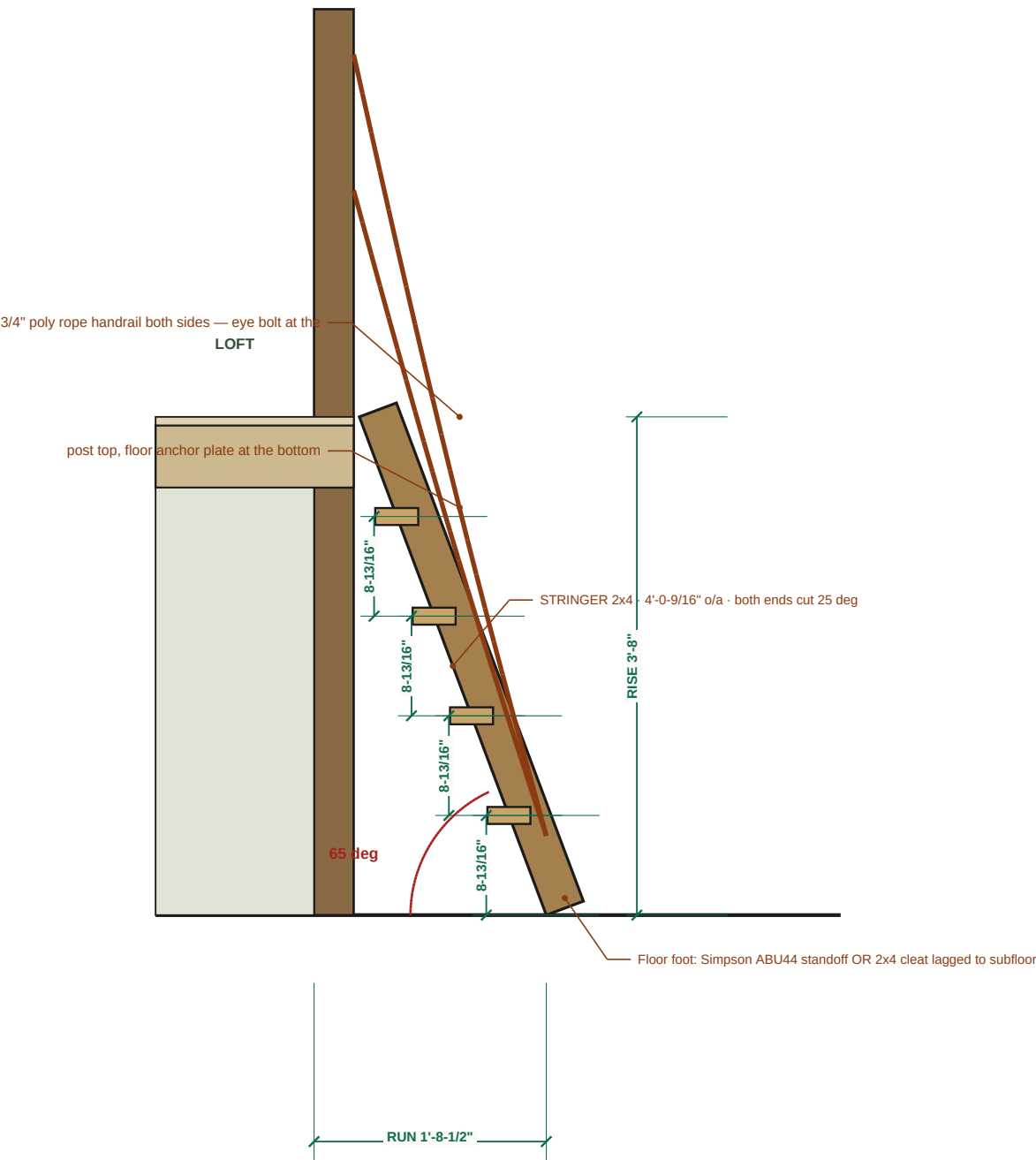
2" grid — scale off the grid, not the paper



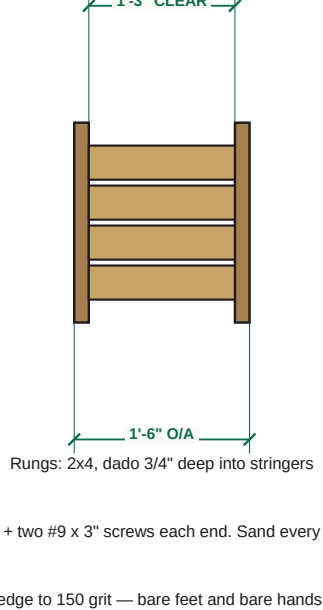
DOOR BUILD-UP

- Slab: two layers of 3/4" ply, glued and screwed, grain crossed, cut to the template above = 1-1/2" thick.
- Face with 1x4 tongue-and-groove pine, vertical, ripped to the arch with a jigsaw.
- Hinges: 12" black strap hinges, 2 per door, through-bolted with 1/4" carriage bolts — screws alone will sag.
- Jamb: 3/4" ply, 3 sides, glued and screwed into the king/jack studs. No sill — remove the bottom plate inside the R.O. after the wall is standing.
- Knob: 2-3/4" round wood knob, 1/4" lag from the back.
- Magnetic catch, not a latch. A kid must be able to push the door open from the inside with no hardware.
- NO lock of any kind, inside or out.
- Paint: deep green (see A9). Two coats, sand between.

LADDER ELEVATION

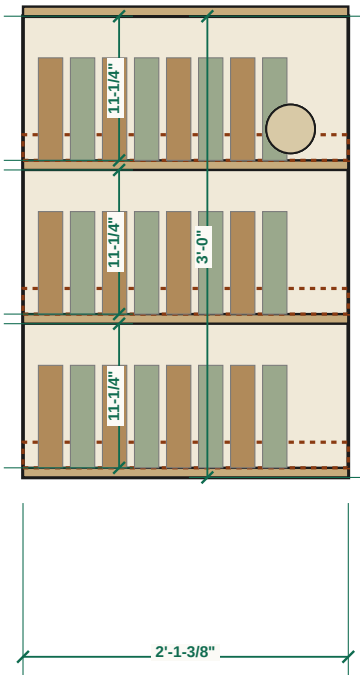


LADDER PLAN



BOOKSHELF GUARD — ELEVATION

fits between posts C and D · 7-1/4" deep



THIS SECTION IS A GUARDRAIL FIRST, A SHELF SECOND.

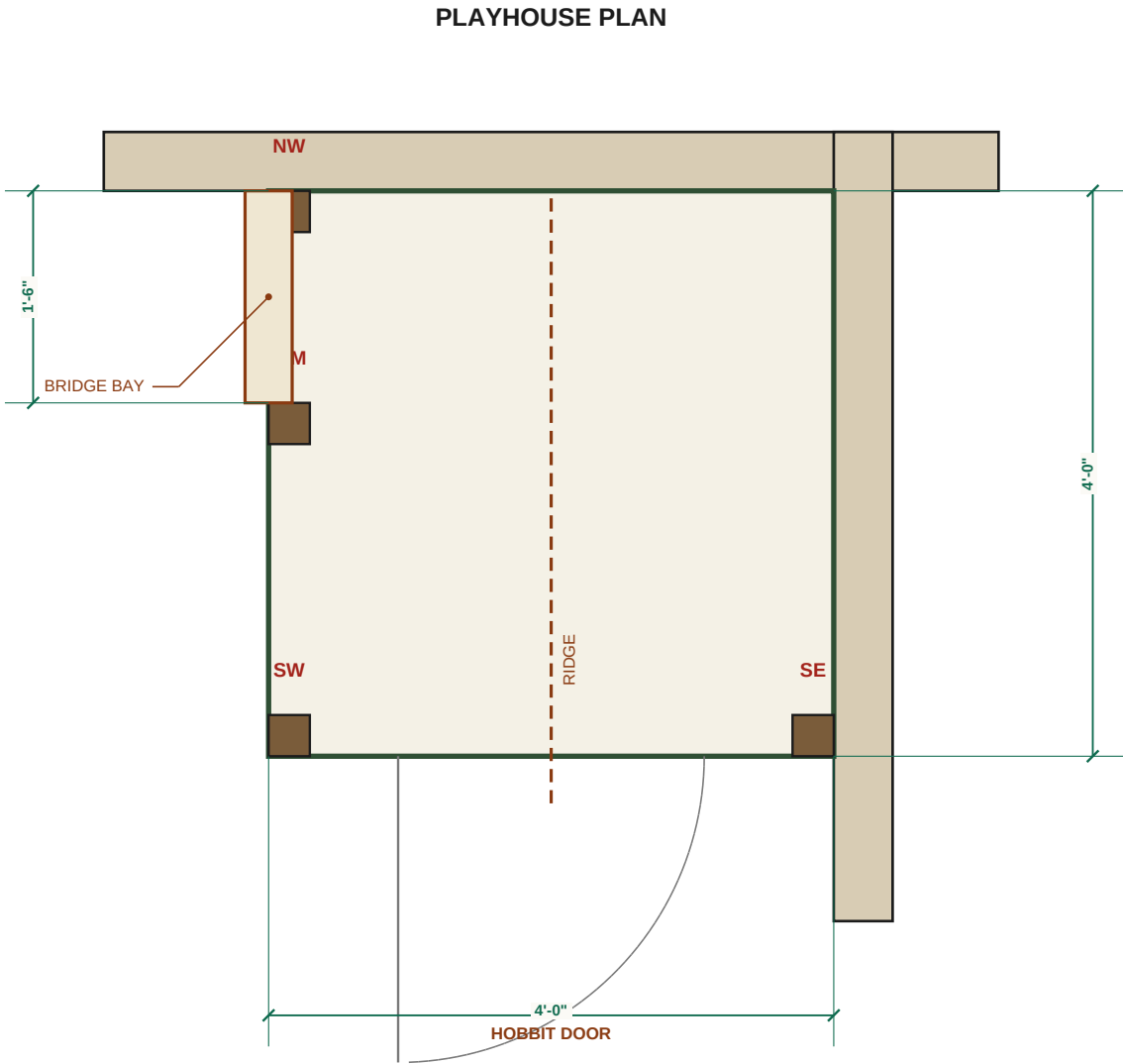
- Carcass: 3/4" ply, dadoed shelves, glued + screwed.
- Lag the two ends to posts C and D with 4x 1/4" x 3" lag screws each — 2 top, 2 bottom. It must not rack.
- 1/4" roundover on every exposed edge, then sand 150.
- 2" book lip on the deck side of every shelf.
- Apothecary jars: HDPE or PET only. No glass, no mason jars. Dried botanicals, acorns, pinecones.
- Hot-glue or silicone the jar bases to the shelf.
- Nothing heavier than 2 lb above 24" off the deck.

PLAYHOUSE

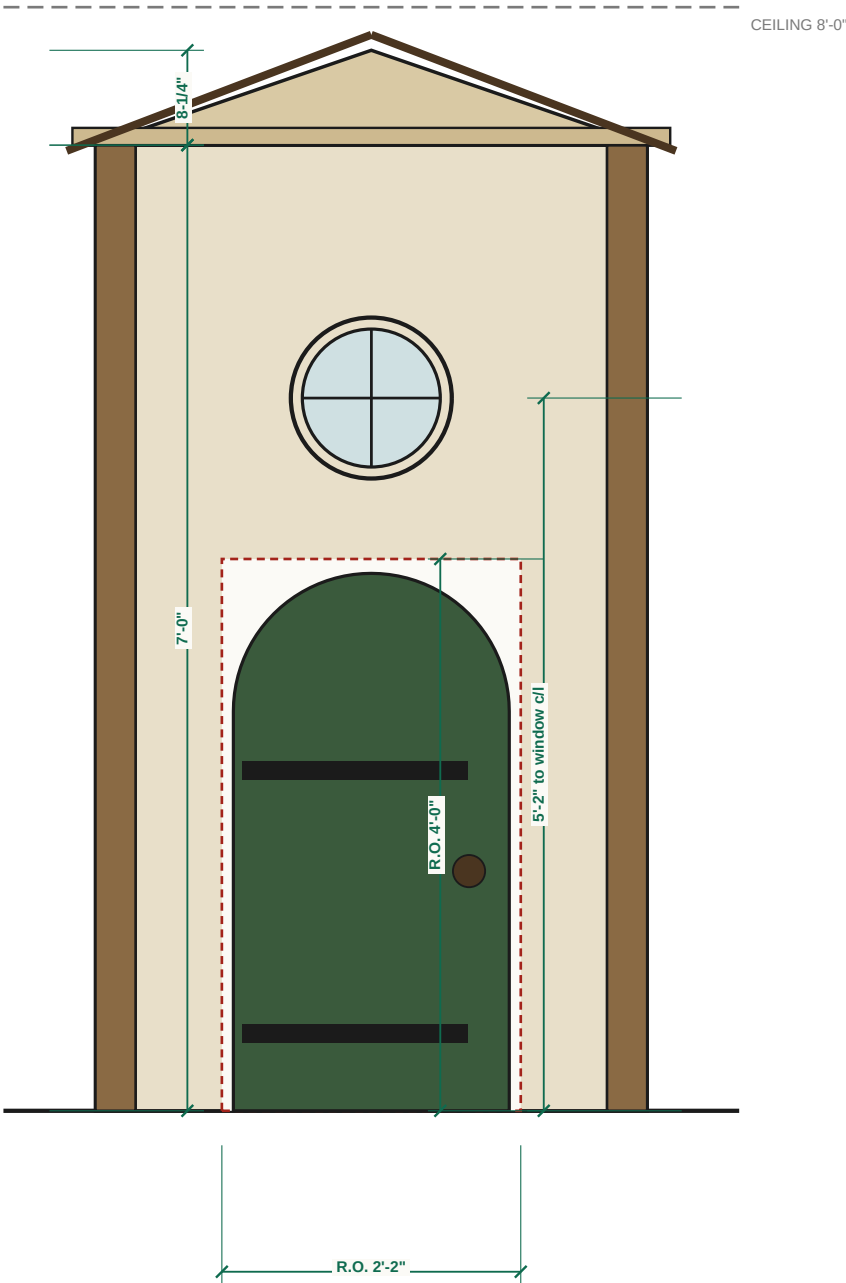
plan · south elevation · roof framing

A6

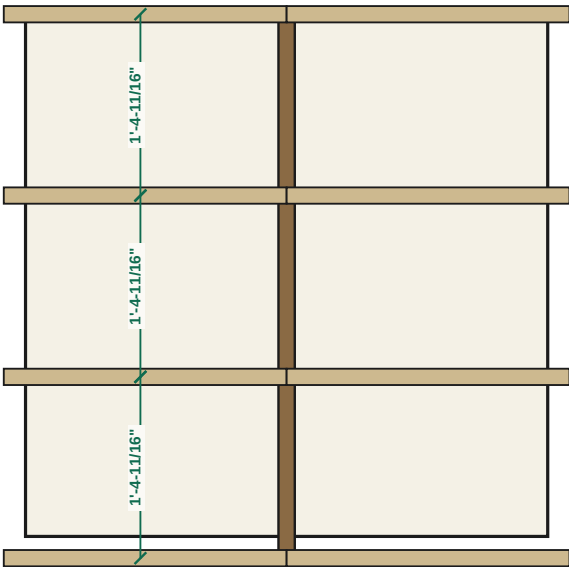
PLAYHOUSE PLAN



SOUTH ELEVATION (faces the room)



ROOF FRAMING PLAN



8 rafters 2x4 · 3.38:12 pitch · ridge 2x6

PLAYHOUSE NOTES

WALLS STOP AT 84", NOT 96".

A 96"-tall box under a 96" ceiling has no room for a roof. Dropping the wall to 84" buys a real 10-1/2" peaked roof that reads from across the room, and still leaves 84" of interior headroom.

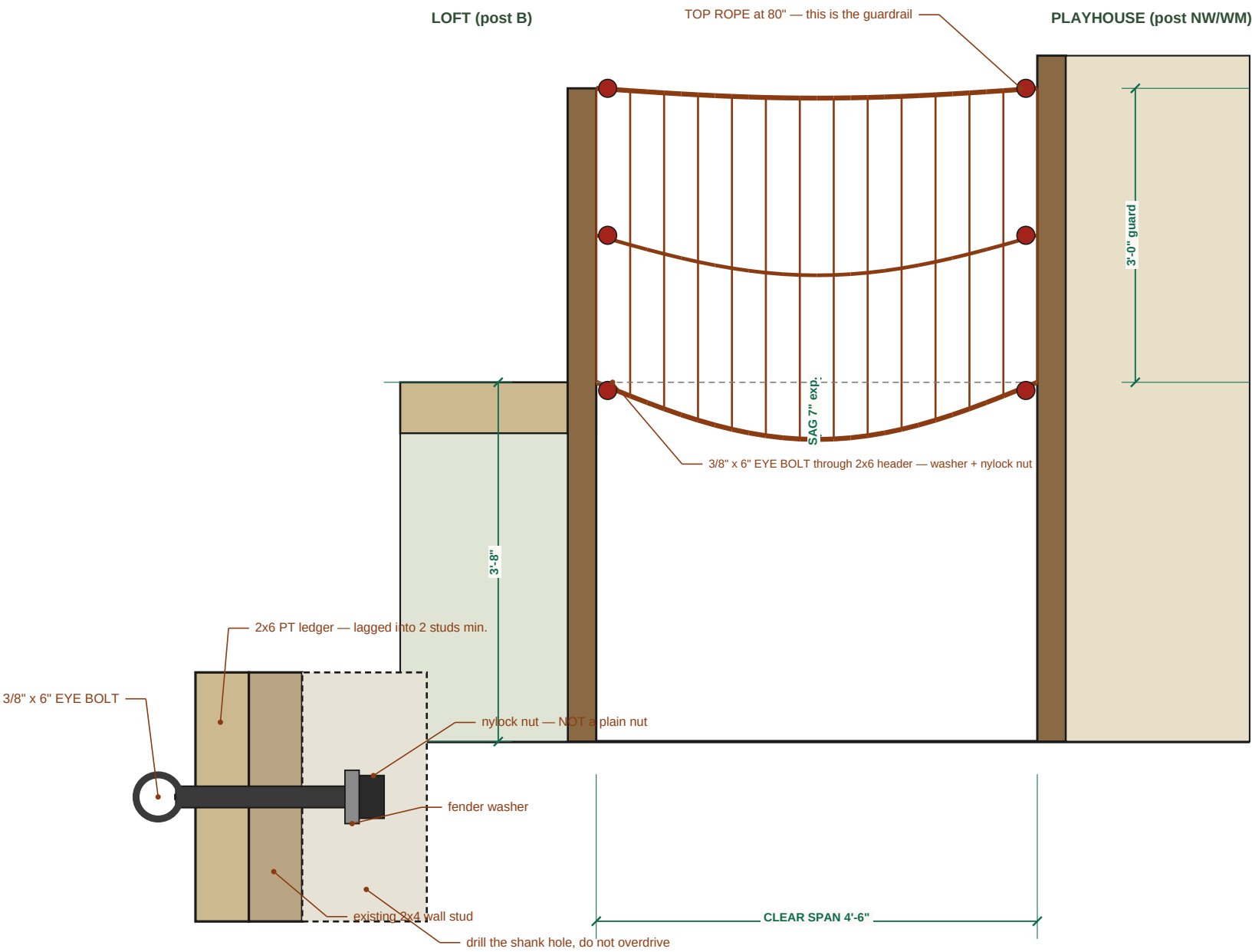
- NW and SE posts lag directly to wall studs: 3x 3/8" x 4" lag + washer each, staggered.
- SW post is free-standing — anchor with a Simpson ABU44Z post base lagged to the subfloor.
- WM post creates the 18" bridge bay and matches the loft gate width. Do not omit it.
- Rope-wrap the SW post full height: 1/2" poly rope, spiral-wound, hot-glued every 6".
- Rough-saw look: run a wire wheel over the 4x4s, then a coat of dark walnut stain wiped back.
- Interior: battery LED lantern on a hook, no wiring.
- Leave the west wall below the bridge bay open or slatted so the space does not feel like a closet.

ROPE BRIDGE

elevation · anchor detail · span correction · safety

A7

BRIDGE ELEVATION — looking north



BRIDGE ANCHOR — enlarged detail

same detail at both ends and at every rope railing anchor

SPAN CORRECTION

The brief says ~42". The room says 4'-6".

13'-6" wall - 5'-0" loft - 4'-0" playhouse
= 4'-6" clear.

Both structures are in corners, so neither can slide toward the other. Order the net for 54", not 42".

SAFETY — READ THIS BEFORE ORDERING ROPE

- A single top rope 36" above a sagging net is NOT a guard. Order a net with full-height SIDE PANELS, or add the mid rope at 18" shown above. Both is better. A kid will lean on whatever is there.
- Commercial cargo net, 1/2" poly or nylon rope, 4"-6" mesh, playground rated. Not hardware-store rope tied by hand. Not climbing rope.
- Eye BOLTS, 3/8" x 6", through-bolted with a fender washer and a nylock nut on the back side. Screw eyes pull out of wood under cyclic load.
- Both ledgers lag into a MINIMUM of 2 studs each, 3/8" x 4" lag screws, 2 per stud.
- Fall zone: the bridge is 44" up. Put 2" interlocking foam or a 1" gym mat on the floor under the full span and 24" past each end. This is not optional.
- Rated for ONE child at a time. Post a sign. Kids will ignore it; the hardware should not.
- Re-torque every eye bolt and lag after 30 days, then every 6 months. Write the date on the ledger.

CUT LIST

every member, every finished length

A8

CUT LIST — generated directly from the model. Every length is the finished cut length.

Lengths are long-point to long-point where a member is mitered. Cut one of any repeated part, test-fit it, then batch the rest.

LOFT / POSTS

STOCK	LENGTH	QTY	NOTE
4x4 PT	3'-1-3/4"	1	room-corner stub
4x4 PT	6'-8"	1	bridge post
4x4 PT	6'-8"	3	
4x4 PT	6'-8"	1	ladder post

LOFT / LEDGERS

STOCK	LENGTH	QTY	NOTE
2x6 PT	5'-0"	1	lag to studs @16 O.C., 3/8 x 4 lag + washer
2x6 PT	4'-10-1/2"	1	lag to studs @16 O.C.

LOFT / RIM

STOCK	LENGTH	QTY	NOTE
2x6	1'-8"	2	outside face length 20.000"
2x6	2'-6-5/8"	2	outside face length 30.615"

LOFT / JOISTS

STOCK	LENGTH	QTY	NOTE
2x6	4'-9"	1	LUS26 hanger both ends
2x6	4'-4"	1	skew-cut; Simpson LSU26 skewable hanger
2x6	3'-9-3/8"	1	skew-cut; Simpson LSU26 skewable hanger
2x6	1'-5"	1	decking support at 135-deg corner

LOFT / KNEE WALL

STOCK	LENGTH	QTY	NOTE
2x4 PT	1'-8"	2	20.00" face length
2x4	1'-8"	2	20.00" face length
2x4	2'-10-3/4"	8	34.75" - NOT 40-42"
2x4 PT	2'-6-5/8"	2	30.61" face length
2x4	2'-6-5/8"	2	30.61" face length

LOFT / HOBBIT DOOR

STOCK	LENGTH	QTY	NOTE
2x4	2'-8"	2	32.00" jack
2x4	2'-10-3/4"	2	34.75" king
2x4	2'-9"	1	flat - deck load bypasses this wall
2x4	2-3/4"	3	2.75"

LADDER

STOCK	LENGTH	QTY	NOTE
2x4	4'-0-9/16"	2	48.55" o/a, 65 deg, both ends cut 25 deg
2x4	1'-3"	4	riser 8.80", dado 3/4" into stringers + 2 screws/end

LOFT / BOOKSHELF

STOCK	LENGTH	QTY	NOTE
3/4" ply	2'-1-3/8"	4	25.39" x 7.25" - 1/4" roundover all edges
3/4" ply	3'-0"	2	lag to post w/ 2x 1/4 x 3 lag
1x3	2'-1-3/8"	3	2" book lip, rounded

PLAYHOUSE / POSTS

STOCK	LENGTH	QTY	NOTE
4x4 PT	7'-0"	1	exposed room corner
4x4 PT	7'-0"	1	bridge bay post
4x4 PT	7'-0"	1	lag to N wall studs
4x4 PT	7'-0"	1	lag to E wall studs

PLAYHOUSE / PLATES

STOCK	LENGTH	QTY	NOTE
2x4	4'-0"	4	top plate @ 84" - anchors roof + trim

PLAYHOUSE / ROOF

STOCK	LENGTH	QTY	NOTE
2x6	4'-2"	1	1x6 or 2x6 ridge, 1" below ceiling
2x4	2'-3"	8	3.38:12 pitch, 2" overhang

PLAYHOUSE / DOOR

STOCK	LENGTH	QTY	NOTE
2x4	4'-0"	2	48" jack
2x4	7'-0"	2	84" king
2x6	2'-9"	1	double 2x6 header over 26" R.O.

BRIDGE / LEDGERS

STOCK	LENGTH	QTY	NOTE
2x6 PT	1'-7-1/2"	2	lag 3/8 x 4 into 2 studs min; eye bolts THROUGH-bolt
2x6	1'-6"	2	net anchor header, 4x 3/8 eye bolts
2x4	1'-6"	2	top rope anchor, 2x 3/8 eye bolts

SHEET GOODS + WHAT TO ACTUALLY BUY

ITEM	NET	BUY
4x4 x 8 ft PT post	64.5 lf	8
4x4 x 10 ft PT post	(1 post + the stub)	1
2x6 x 8 ft Ground Contact PT	13.1 lf	3
2x6 x 8 ft SPF	32.6 lf	5
2x4 x 8 ft Ground Contact PT	8.4 lf	2
2x4 x 8 ft SPF stud	118.3 lf	18
1x3 x 8 ft pine (book lips)	6.3 lf	1
3/4" 4x8 ply — deck + shelf	21.75 sq ft + shelf	2
3/4" 4x8 ext ply — skins + doors	~46 sq ft	2
1x4 T&G pine — door faces	~24 lf	1 bdl

2x quantities include 15% waste. Every cut in this set fits an 8 ft stick.
The one 10 ft 4x4 yields an 80" post AND the 37-3/4" corner stub with almost no waste. Nine 8-footers would leave you one post short.

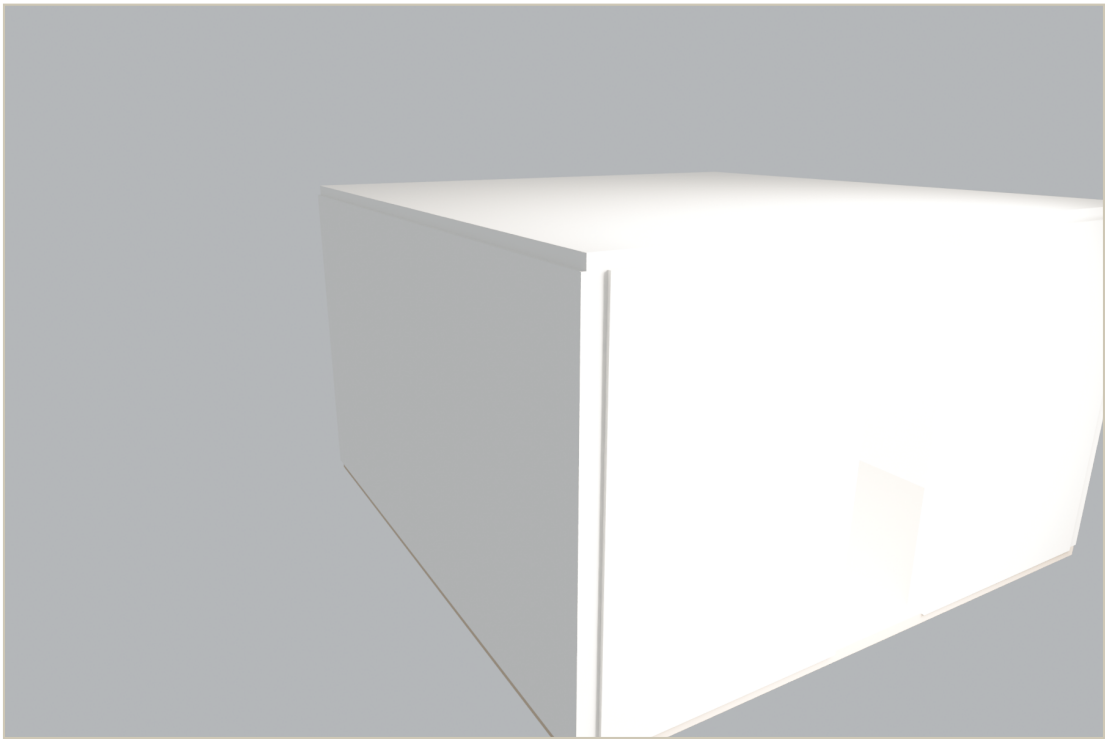
HARDWARE SCHEDULE + HOME DEPOT BUY LIST

Internet # verified against homedepot.com. Prices are US national ranges and move weekly — treat them as a budget, not a quote. Items without an Internet # are commodity stock carried in every store; grab them off the shelf.

ITEM	INTERNET #	QTY	EST. EACH	WHAT IT DOES / WHY THIS ONE
LUMBER				
4 in. x 4 in. x 8 ft. #1 Pressure-Treated Post	202812131	8	\$12 - \$16	four loft posts at 80" + four playhouse posts at 84"
4 in. x 4 in. x 10 ft. #1 Pressure-Treated Post	—	1	\$16 - \$22	yields the fifth 80" post AND the 37-3/4" corner stub
2 in. x 6 in. x 8 ft. Ground Contact PT	—	3	\$9 - \$13	ledgers: loft N + W, bridge both sides
2 in. x 6 in. x 8 ft. Prime Kiln-Dried SPF	—	6	\$7 - \$10	joists, rim, playhouse header, ridge
2 in. x 4 in. x 8 ft. Ground Contact PT	—	2	\$6 - \$9	knee wall bottom plates (on subfloor)
2 in. x 4 in. x 8 ft. Prime Kiln-Dried Stud	—	18	\$4 - \$6	studs, plates, rafters, ladder, door framing
1 in. x 3 in. x 8 ft. Pine Board	—	1	\$6 - \$9	shelf book lips
3/4 in. 4x8 Sanded Plywood (BC or Sande)	—	4	\$48 - \$68	2 structural, 2 skin + doors
1 in. x 4 in. Tongue & Groove Pine (bundle)	—	1	\$28 - \$40	door faces, both doors
STRUCTURAL HARDWARE				
Simpson Strong-Tie LUS26 Face-Mount Joist Hanger	100374889	8	\$1.77 ea	square joist-to-ledger connections
Simpson Strong-Tie LSU26 Adjustable Skewable U Hanger	100375129	6	\$4 - \$6 ea	J2/J3 into the angled rims — THE part for this
Simpson Strong-Tie ABU44Z Adjustable Standoff Post Base	100375358	4	\$14 - \$19 ea	free-standing posts: loft D, playhouse SW, ladder foot
Simpson Strong-Tie A23 / A35 Framing Angle	—	16	\$0.80 ea	post-to-rim and plate ties
Everbilt 3/8 in. x 4 in. Hex Lag Screw (bulk)	—	60	\$0.55 ea	ledgers into studs, posts into studs
Everbilt 3/8 in. Flat Washer (bulk)	—	100	\$0.15 ea	every lag and every bolt, both sides
Everbilt 3/8 in. x 6 in. Zinc-Plated Eye Bolt w/ Nut	806746	12	\$1.11 ea	bridge + rope railing anchors — BOLTS not screw eyes
Everbilt 3/8 in. Nylon Insert Lock Nut (nylock)	—	16	\$0.35 ea	back side of every eye bolt
Everbilt 1/4 in. x 3 in. Hex Lag Screw	—	40	\$0.40 ea	bookshelf to posts, hinges, trim
#9 x 3 in. Star Drive Construction Screw, 5 lb box	—	1	\$34 - \$44	general framing
#8 x 1-5/8 in. Construction Screw, 1 lb box	—	2	\$9 - \$12	decking + skins — SCREWED, never nailed
ROPE, NET, PLAY HARDWARE				
VEVOR Climbing Cargo Net 6.6 x 10.5 ft Double-Layer Rope Bridge	320569708	1	\$70 - \$110	cut/fold to 54 x 18 — double layer gives side panels
VEVOR Playground Climbing Cargo Net 8 x 4 ft	320580952	1	\$45 - \$70	ALTERNATE if you want the smaller single panel
Everbilt 1/2 in. x 50 ft. Twisted Poly Rope (manila look)	—	2	\$22 - \$32	guard ropes, ladder handrails, post wrap
Everbilt 1/2 in. Rope Thimble + Wire Rope Clip set	—	16	\$2 - \$4 ea	every rope termination — no knots at anchors
Everbilt 3/8 in. Quick Link / Screw-Pin Shackle	—	8	\$3 - \$5 ea	net to eye bolt, serviceable
DOORS, FINISH, LIGHT				
NUVO IRON 12 in. Black Steel Strap Hinges Kit (2-pack)	320202663	2	\$18 - \$26	one 2-pack per door
2-3/4 in. Round Wood Cabinet Knob	—	2	\$4 - \$7	hobbit door knobs
Everbilt Magnetic Push Catch	—	2	\$4 - \$6	NO latch, NO lock — kid must push out from inside
8 in. Clear Acrylic Circle / Plexiglass sheet to cut	—	1	\$12 - \$22	porthole — acrylic only, never glass
12 in. Clear Acrylic Circle	—	1	\$16 - \$28	playhouse round window
Commercial Electric LED Puck Light, battery, 3-pack	—	2	\$18 - \$26	2 inside the loft base, 1 in the playhouse
BEHR Premium Plus Interior Eggshell — deep green	—	1 qt	\$20 - \$26	match "Dark Everglade" / "Pine Mountain"
Varathane Dark Walnut Wood Stain, qt	—	1	\$14 - \$19	rough-sawn look on 4x4s and trim
Minwax Polycrylic Clear Satin, qt	—	1	\$20 - \$26	water-based topcoat — low odor for a kid room
Zero-VOC primer, qt	—	1	\$16 - \$22	everything a kid touches
FALL PROTECTION — NOT OPTIONAL				
3/4 in. Interlocking Foam Mat Tiles, 24x24, 6-pack	—	3	\$26 - \$38	floor under the bridge and both entries

BUDGET: \$1,050 - \$1,350 in materials. Add ~\$120 if you buy the double-layer net instead of the single panel.

MODEL VIEWS — these are the geometry, not a mood board. Use them to check that what you build matches what was drawn.



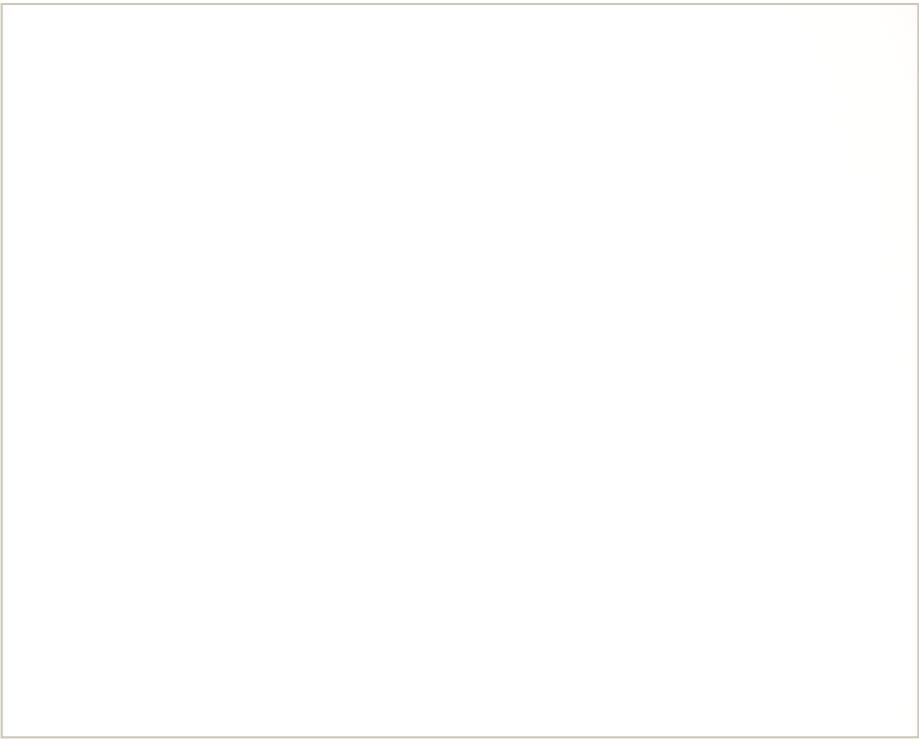
01 Isometric overview — both structures and the crossing



02 From the room doorway — what a kid sees walking in

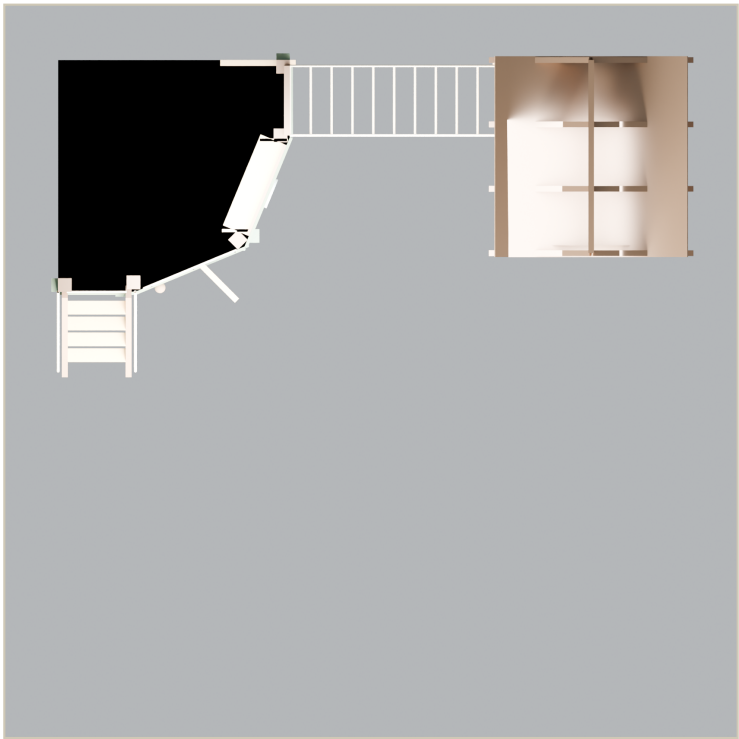


03 Loft three-quarter — hex deck, rope guard, bookshelf



04 Playhouse front — hobbit door, round window, peaked roof

ORTHOGRAPHIC + FRAMING-ONLY VIEWS



T1 True plan — room shell removed



T2 Loft framing only — skins, deck and rope hidden



T3 North elevation



T4 West elevation